Exp 4: Setting Up a Local Area Network

Aim:

To design, setup and configure a LAN connecting 4 components through a network switch, assign static IP address to all devices, verify connectivity using ping command and demonstrate resource or data sharing across network

Equipments Required:

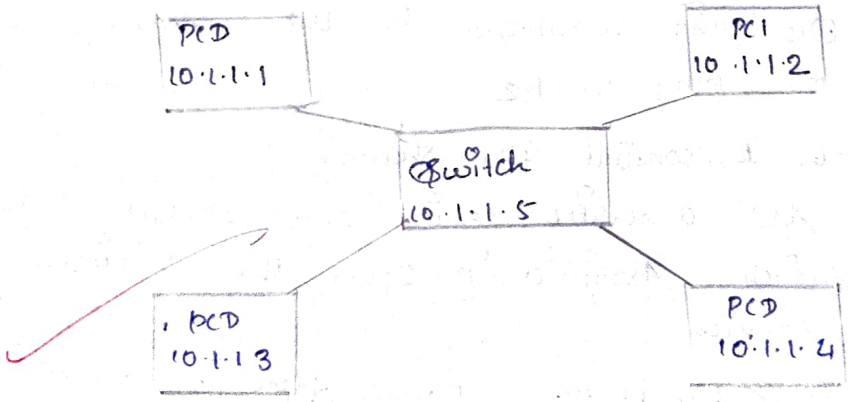
1)	Personal Computer (PC)	4	Ethernet (NIC) enabled
2)	Network Switch	1	8/16/24 port unmanaged / managed
3)	Ethernet Cables Coarse / Cat6)	4	Straight-through RJ-45 connectors

Theory:

⇒ LAN is a network that connects computer & devices within limited area like home, Office...

⇒ Star Topology → all end devices are individually connected to central switch

Network Diagram:



IP	Address	Configuration	Table
PC0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa 0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation of Cisco Packet Tracket

physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software. following the steps below

Step 1: Place the Devices

Open cisco packed tracer. Seled 4 PC's onto workspace. seled & drag & one switch.

Step 2: Cable the Devices

click connection → copper straight wire and connect all 4 PCs to switch via Fast Ethernet port

Step 3: Assign IP Address to PC's

Step 4: click the switch → CLI tab. Enter following commands

→ Enable
→ configure terminal
→ interface vlan1
→ ip address 10.1.1.5 255.0.0.0
→ no shutdown
→ exit

Step 5: Verify connectivity using Ping

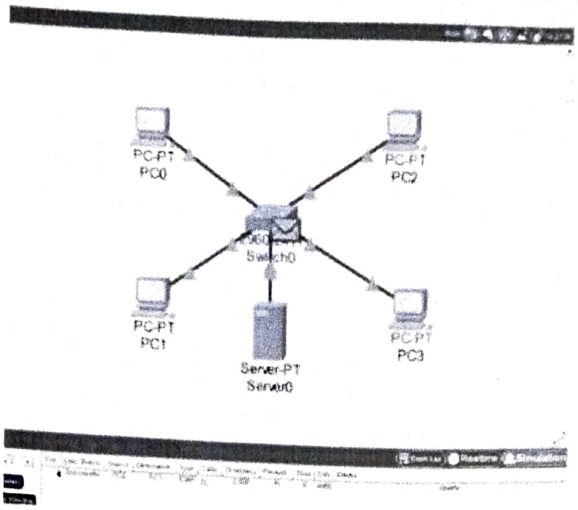
click Pc → Desktop → Command propt
Use ping command to test connectivity
eg: ping 10.1.1.2

Step 6: Demonstrate Data sharing

Add a server device and connect it to switch. Assign it ip. Open the server service.

Step 7: Observe the packet flow in simulation mode

Switch to simulation mode send a ping request btwn two device. capture the step through packet as it travel hop by hop through Switch. Observing mac address based forwarding



Note

Result :

Thus, setting up a Local Area Network is implemented successfully.